

0 8

A scientist investigated birth mass in a population of babies. She determined the birth mass (b) of babies and grouped this information into different ranges of birth mass.

Her results are shown in **Table 4**.

Table 4

Birth mass b / kg	Range of mass / kg	Frequency density
$0.0 < b \leq 2.0$	2.0	5 000
$2.0 < b \leq 2.5$	0.5	20 000
$2.5 < b \leq 3.0$	0.5	90 000
$3.0 < b \leq 3.5$	0.5	260 000
$3.5 < b \leq 4.5$	1.0	200 000
$4.5 < b \leq 5.5$	1.0	20 000

Frequency density is calculated using this equation

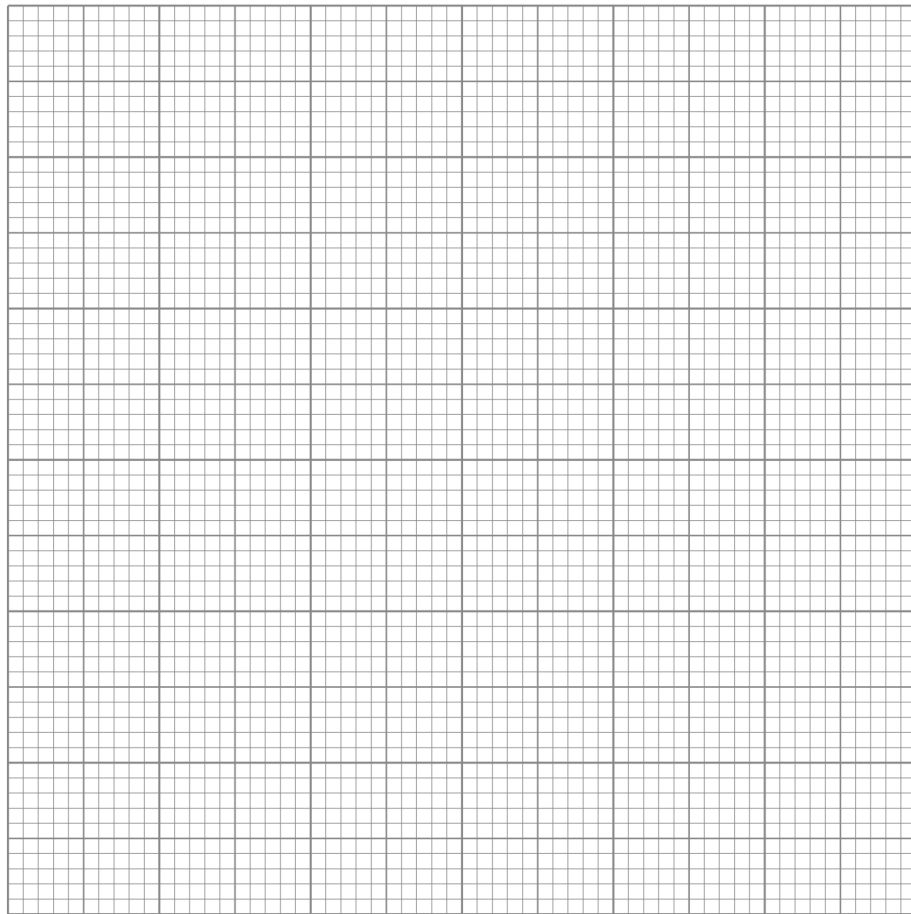
$$\text{Frequency density} = \frac{\text{number of babies}}{\text{range of mass}}$$

Do not write
outside the
box



0 8 . 1

Draw, on **Figure 8**, a **suitable** chart to show the distribution of birth mass for this population of babies.

[4 marks]**Figure 8**Frequency
density

Birth mass / kg

0 8 . 2

Babies with birth mass less than 2.5 kg are classified as low birth mass.

Use information in **Table 4** and the equation to calculate the number of babies born with low birth mass in this population.

Show your working.

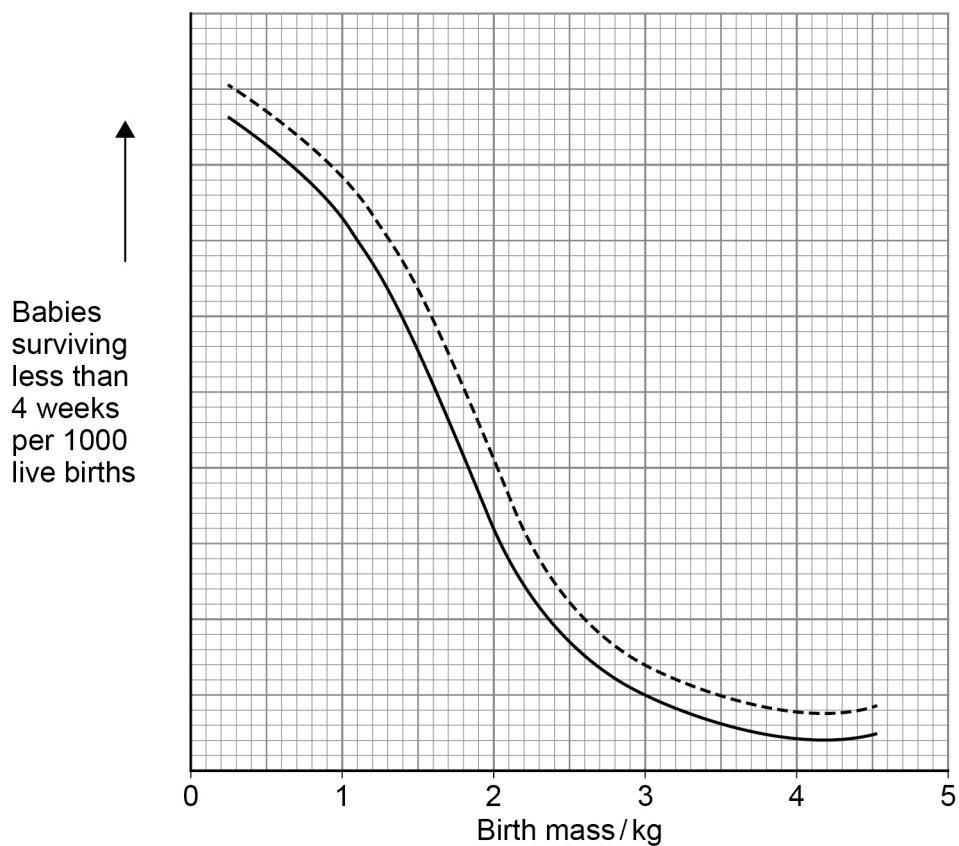
[2 marks]

Answer _____

Question 8 continues on the next page**Turn over ►**

The scientist also measured the relationship between birth mass and babies surviving less than 4 weeks. She determined if the mothers of these babies smoked cigarettes during pregnancy. Her results are shown in **Figure 9**.

Figure 9



Key

- Mothers who smoked cigarettes during pregnancy
 ————— Mothers who did not smoke cigarettes during pregnancy

0 8 . 3 State **three** conclusions that can be drawn from the data in **Figure 9**.

[3 marks]

- 1 _____
- _____
- _____
- 2 _____
- _____
- _____
- 3 _____
- _____
- _____

